

7. Barnacles are small animals that grow on rocks at the seashore. Two species of barnacle native to the British Isles, *Chthamalus stellatus* and *Semibalanus balanoides* are shown in the photographs in **Image 7.1**.

Image 7.1



C. stellatus



S. balanoides

Students carried out a survey to test the following hypothesis:

- C. stellatus* is found further up the seashore towards the high water mark.
S. balanoides is found lower down the seashore towards the low water mark.

Image 7.2



Photograph of students carrying out their survey



The students measured the abundance and distribution of barnacles between the high water mark and the low water mark on the seashore. The results of their survey are shown in **Table 7.3**.

Table 7.3

Distance from high water mark (m)	Number of barnacles	
	<i>C. stellatus</i>	<i>S. balanoides</i>
0 (high water mark)	0	0
5	13	0
10	28	0
15	36	0
20	52	3
25	57	25
30	29	41
35	17	51
40	0	60
45	0	42
50 (low water mark)	0	0

- (a) Describe the method the students would have used to measure the abundance and distribution of barnacles between the high and low water mark. [3]

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(b) (i) State **two environmental** factors which could affect the population of barnacles. [2]

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(ii) Calculate the ratio of *C. stellatus* to *S. balanoides* at the **25 m** mark. [1]
Space for working.

C. stellatus : *S. balanoides*

(iii) State the type of competition occurring between the two species of barnacles at the 25 m mark. [1]

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(iv) State whether the results of the survey support the students' hypothesis (given on page 15). Use the data provided in **Table 7.3** to give a reason for your answer. [1]

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